

Dataset

Data access

The olive flounder image dataset is not distributed with this repository. As stated in the manuscript, data may be requested from the corresponding authors and are provided subject to the applicable access and use conditions.

Cohort and split

The dataset contains 1,461 RGB images of *Paralichthys olivaceus*, including ocular- and blind-side views. The split is made by individual fish: every view of one fish remains in a single split.

Split	Images	Fish
Training	1,169	640
Validation	292	160
Total	1,461	800

The proposed method and all comparison models use the same split.

Symptom classes

Counts are non-exclusive because one image can contain more than one visible symptom.

ID	Symptom class	Images	Annotations	Route assignment	Allowed part
1	Body lesion	324	513	Body	Body
2	Mouth ulcer	326	328	Mouth	Mouth
3	Fin deformity	711	1,089	Fin	Fin
4	Fin necrosis	215	308	Fin	Fin
5	Fin base necrosis	491	571	Fin	Body or fin
6	Caudal deformity	283	314	Caudal fin	Caudal fin
7	Caudal necrosis	117	128	Caudal fin	Caudal fin
8	Caudal base necrosis	85	85	Caudal fin	Body or caudal fin

Background is class 0. The class order is fixed in [configs/progressive_candidate_correction.yaml](#) and [data/contract.py](#).

Directory and annotation format

```
dataset-root/  
  annotations/  
    train_annotations.json  
    val_annotations.json  
  train/  
    images/  
    masks/  
    part_masks/  
  val/  
    images/  
    masks/  
    part_masks/
```

The two annotation files use COCO-style **images**, **annotations**, and **categories** arrays. Symptom regions may be stored as polygons or RLE. The loader rasterizes them into a nine-class semantic target; where annotated regions overlap, the smaller region is assigned last and therefore retains its class label.

Each **images** entry contains **id**, **file_name**, **width**, **height**, and a pseudonymous **fish_id**. Image names are shared by the corresponding PNG files in **masks** and **part_masks**.

- **masks/<stem>.png** is the binary Fish Region.
- **part_masks/<stem>.png** is the five-class Part Map: background, body, fin, caudal fin, and mouth.
- Semantic masks are generated from the COCO annotations and cached under **.cache/semantic_masks**.

Part Maps were generated by a separate segmentation process, structurally refined, and visually reviewed. They supervise Part-structure information and are used for allowed-part analysis. Inference by the proposed network requires only RGB input. The part-assisted SegFormer comparison receives the reviewed Part Map as an additional input and is interpreted as a reference-input comparison.

Preprocessing

Training and evaluation use 384×384 inputs with ImageNet mean and standard-deviation normalization. RGB images use bilinear resizing; semantic targets, Fish Regions, and Part Maps use nearest-neighbor resizing.

Spatial augmentation is applied jointly to the RGB image and all masks before the dense targets in [SUPERVISION.pdf](#) are derived. Depth is not used.

Dataset validation

Run the supplied checker before training:

```
python scripts/verify_dataset.py --data-root /path/to/dataset-root
```

The checker verifies split counts, required image/mask pairs, valid class IDs, Fish Region–Part Map agreement, and separation of **fish_id** values between the training and validation splits.

Candidate evaluation

Candidate evaluation first resizes each semantic target to 384×384 and then forms connected components independently for every symptom class. One 8-connected foreground region is one image-local component. Under the study split and annotation setting this produces 1,064 components.

A symptom region is captured when at least 50% of its pixels are selected. For each class and evaluated response, the highest threshold reaching at least 70% symptom-region capture on the training set is fixed and applied unchanged to the validation set. Allowed-part agreement rate is computed from class-specific candidate pixels outside all symptom annotations. Outside-annotation candidate area ratio uses the union of class-specific candidates inside the Fish Region, with Fish Region area as its denominator.

The rasterization size, connectivity, coverage criterion, and split are part of the evaluation protocol because changing any of them changes the evaluation unit.